Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 35)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 1 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 1 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 2 . 1 Explain how a ground-source heat pump can deliver more heat energy to a building than the electrical energy used to run it.

[3 marks]

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0 2 . 2 Explain how Concentrating Solar Power (CSP) with thermal storage overcomes the intermittency of solar power.

[3 marks]

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0 2 . 3 Explain why CSP can only be used in areas with little cloud, whereas PV can be used anywhere.

[2 marks]

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0 2 . 4 Describe **two** environmental impacts of solar PV.

[2 marks]

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0 3 A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

0 3 . 1 Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation.

[2 marks]

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0 3 . 2 Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

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0 3 . 3 Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

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0 4 Table 1 shows some properties of four groups of insecticide.

Table 1

Insecticide group	Toxicity to insects	Toxicity to mammals	Persistence	Liposolubility
Organochlorines (eg DDT)	High	Low	High	High
Organophosphates (eg malathion)	High	High	Low	Low
Pyrethroids	High	Low	Low	Low
Neonicotinoids	High	Low	Moderate	Low (water soluble)

0 4 . 1 Use Table 1 to explain why organochlorine insecticides caused the decline of birds of prey such as the peregrine falcon in the 1950s and 1960s.

[4 marks]

0 4 . 2 Organophosphates replaced organochlorines for many uses. Use Table 1 to give **one** advantage and **one** disadvantage of this change.

[2 marks]

Advantage

Disadvantage

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0 4 . 3 Explain why pyrethroids should not be used near rivers or fish farms.

[1 mark]

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0 4 . 4 Describe how the properties of neonicotinoids allow them to be used as a seed coating, and explain why this method of application reduces harm to non-target species.

[3 marks]

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0 5 . 1 Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

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0 5 . 2 Describe the process of phytomining.

[3 marks]

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0 5 . 3 Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

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0 6 Table 1 shows three specialist solid wastes.

0 6 . 1 Complete Table 1 to show the main hazard of each waste and the disposal method used.

[6 marks]

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

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0 6 . 2 Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

0 6 . 3 Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

0 6 . 4 Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

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0 7 Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

Table 1

Reservoir	Mass of carbon / billion tonnes
Atmosphere (CO_2)	700
Living organisms	470
Dead organic matter (soil)	3 700
Surface ocean waters	500
Deep ocean (dissolved)	35 000
Fossil fuels	10 000
Sedimentary carbonate rocks	20 000 000

0 7 . 1 Express the mass of carbon in the atmosphere in standard form, in tonnes.

[1 mark]

Answer ☐ tonnes

0 7 . 2 Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

[2 marks]

Answer ☐ %

0 7 . 3 Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO_2 concentrations.

[2 marks]

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0 7 . 4 Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

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0 7 . 5 Explain why aerobic respiration returns carbon to the atmosphere as CO_2 but anaerobic respiration returns it mainly as methane.

[2 marks]

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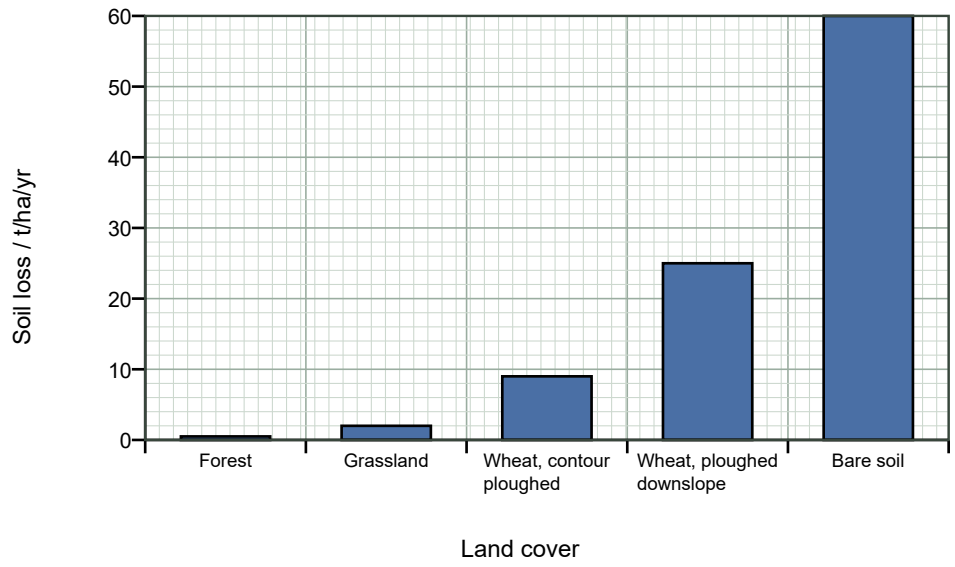
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0 8 Figure 1 shows the annual soil loss measured on experimental plots with different vegetation cover on a 10% slope in a region with heavy seasonal rain.

Figure 1



0 8 . 1 Describe the pattern shown in Figure 1.

[2 marks]

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0 8 . 2 Explain **four** ways in which vegetation reduces the rate of soil erosion.

[4 marks]

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0 8 . 3 Explain why ploughing up and down a slope causes more erosion than contour ploughing.

[2 marks]

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0 8 . 4 Explain why the region's climate makes the soil vulnerable to **both** wind and water erosion.

[2 marks]

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0 9 . 1 Give **three** sources of mercury pollution.

[3 marks]

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0 9 . 2 Describe **three** methods used to control mercury pollution.

[3 marks]

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0 9 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 9 . 4 State what is meant by synergism and give an example involving a heavy metal.

[2 marks]

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1 0 . 1 Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

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1 0 . 2 Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

11 . 2 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS